- **Data Science:**

Data Science is an interdisciplinary field that uses statistics, mathematics, programming, and domain knowledge to collect, analyze, and interpret **data** in order to extract meaningful insights and make better decisions

Data Science is the study of data (raw facts or figures) to extract useful insights and knowledge.

Data Science is a broader field, It covers the entire process.

1. Data collection.
2. Cleaning and preparing data.
3. Analyzing/Exploration and visualizing data.
4. Drawing insights and making decisions.
5. Applying Machine Learning models (if needed)

Machine Learning is a subset *AI*, but used as a tool in data science to learning from data.

1. It uses algorithms and models that learns from data and make predictions or classifications.
2. Examples: predicting house prices, classifying spam emails.

Data Science without ML:

Analyzing sales data to find which product sells best in winters.

Just statistics + visualization, no ML needed.

Data Science with ML:

Using past sales data to predict future sales.

Here ML is applied to make forecasts.

"All Machine Learning is the part of Data Science, but not all data science requires machine learning."

- **YouTube Example: (Data Science vs Machine Learning)**

When YouTube recommend videos, it is not just Data Science but ML is also involved.

1. Data Science:

- Collects and analyzes data (your watch history, likes, comments, subscriptions and watch time) .
- Finds patterns in user behavior.

2. Machine Learning:

- Uses ML models to learn from data and make prediction on what you might like next.

If YouTube only showed Trending videos → that is mostly Data Science (analysis + visualization).

But because it shows personalized recommendations for each user → that is DS + ML.

- **Weather Example: (Data Science vs Machine Learning)**

Data Science:

Meteorologist collects weather data → temperature, humidity, rainfall, wind speed etc.

Use Data Science tools (Pandas, Matplotlib, Seaborn) to analyze and visualize:

1. Average rainfall in August.
2. Which month is hottest.
3. Past weather trends.

This answers “What has happened?” and “What is happening now?”

DS + ML:

Past weather data is fed into Machine Learning models.

The ML model learns patterns from history.

It predicts tomorrow’s weather:

Will it rain tomorrow? What will be the temperature next week?

This answers “What will happen?”

Weather reports (yesterday and today’s data) = Data Science.

Weather forecast (tomorrow’s prediction) = Data Science + Machine Learning.

- **Statistics (Backbone of Data Science):**

Definition: Statistics is the branch of mathematics that deals with collecting, analyzing, interpreting, presenting, and organizing data.

Goal: To understand data and make informed decisions.

Types of Statistics:

1. Descriptive Statistics (What has happened?)
   - Summarizes and describes features of a dataset.
   - Uses numbers, tables, or graphs to present data.
   - Example: *Mean, Median, Mode, Variance, Standard Deviation*.
2. Inferential Statistics (What might happen or what does the data suggest?)
   - Uses a sample of data to make conclusions or predictions about a larger population.
   - Example: *Hypothesis testing, Confidence intervals, Regression.*

1. Descriptive Statistics:

- Definition: Descriptive statistics is the branch of statistics that deals with summarizing and describing data in a meaningful way.
- It does not make predictions; it only tells us what the data shows.
- Examples:
  - Average score of a class = Mean.
  - Middle salary of employees = Median.
  - Most common shoe size = Mode.
  - Spread of test scores = Variance/Standard Deviation.

i. Mean (Average)

- Definition: The sum of all values divided by the total number of values.
- Formula: $$\text{Mean} = \frac{\text{Sum of values}}{\text{Number of values}}$$
- Example: Scores = [10, 20, 30] → Mean = (10+20+30)/3 = 20.

ii. Median (Middle Value)

- Definition: The middle value when data is arranged in order.
- If data count is odd → middle value.
- If data count is even → average of two middle values.
- Example: [10, 20, 30] → Median = 20; [10, 20, 30, 40] → Median = (20+30)/2 = 25.

iii. Mode (Most Frequent Value)

- **Definition:** The value that occurs most often in the dataset.
- **Example:** [10, 20, 20, 30] → Mode = 20.

iv. Variance (Measure of Spread)

- **Definition:** Shows how far values spread out from the mean.
- Higher variance = values are more spread out.
- Formula:

$$\text{Variance} = \frac{\sum (x_i - \text{Mean})^2}{n}$$

- **Example:** If all students score the same, variance = 0

v. Standard Deviation (SD)

- Definition: The square root of variance.
- Easier to interpret because it's in the same units as the data.
- Small SD = values are close to mean, Large SD = values are spread out.
- Example: If average score = 70 and SD = 2, most scores are close to 70.